

Oral Semaglutide and Cardiovascular Outcomes in High-Risk Type 2 Diabetes

Critical Appraisal of the SOUL Trial

Department of Medicine, Santosh Medical College Ghaziabad

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Y. Bi,^{1,2} M. Li,^{1,2} Y. Liu,³ T. Li,⁴ J. Lu,^{1,2} P. Duan,⁵ F. Xu,⁶ Q. Dong,⁷ Ailiang Wang,⁸ T. Wang,^{1,2} R. Zheng,^{1,2} Y. Chen,^{1,2} M. Xu,^{1,2} X. Wang,⁹ Xinhuan Zhang,¹⁰ Y. Niu,¹¹ Z. Kang,¹² C. Lu,¹³ Jing Wang,¹⁴ X. Qiu,¹⁵ An Wang,¹⁶ S. Wu,^{1,2,17} J. Niu,^{1,2,18} Jingya Wang,¹⁹ Z. Zhao,^{1,2} H. Pan,²⁰ X. Yang,²¹ X. Niu,²² S. Pang,²³ Xiaoliang Zhang,²⁴ Y. Dai,²⁵ Q. Wan,²⁶ S. Chen,²⁷ Q. Zheng,²⁸ S. Dai,²⁹ J. Deng,^{1,2} L. Liu,³⁰ G. Wang,³¹ H. Zhu,³² W. Tang,³³ H. Liu,³⁴ Z. Guo,³⁵ G. Ning,^{1,2} J. He,³⁶ Y. Xu,^{1,2} and W. Wang,^{1,2} for the BPROAD Research Group*

ABSTRACT

BACKGROUND

Effective targets for systolic blood-pressure control in patients with type 2 diabetes are unclear.

METHODS

The authors' full names, academic degrees, and affiliations are listed at the end of the article. Drs. Ning, Y. Xu, and W. Wang can be contacted at gning@sibs.ac.cn, jane.wang@cityu.edu.cn, and wangwu@chemu.edu.cn.

Trial Identity and Citation

TRIAL NAME & CITATION

BPROAD Trial: Oral Semaglutide and Cardiovascular Outcomes in High-Risk Type 2 Diabetes

Citation: N Engl J Med 2025;392:2001-12

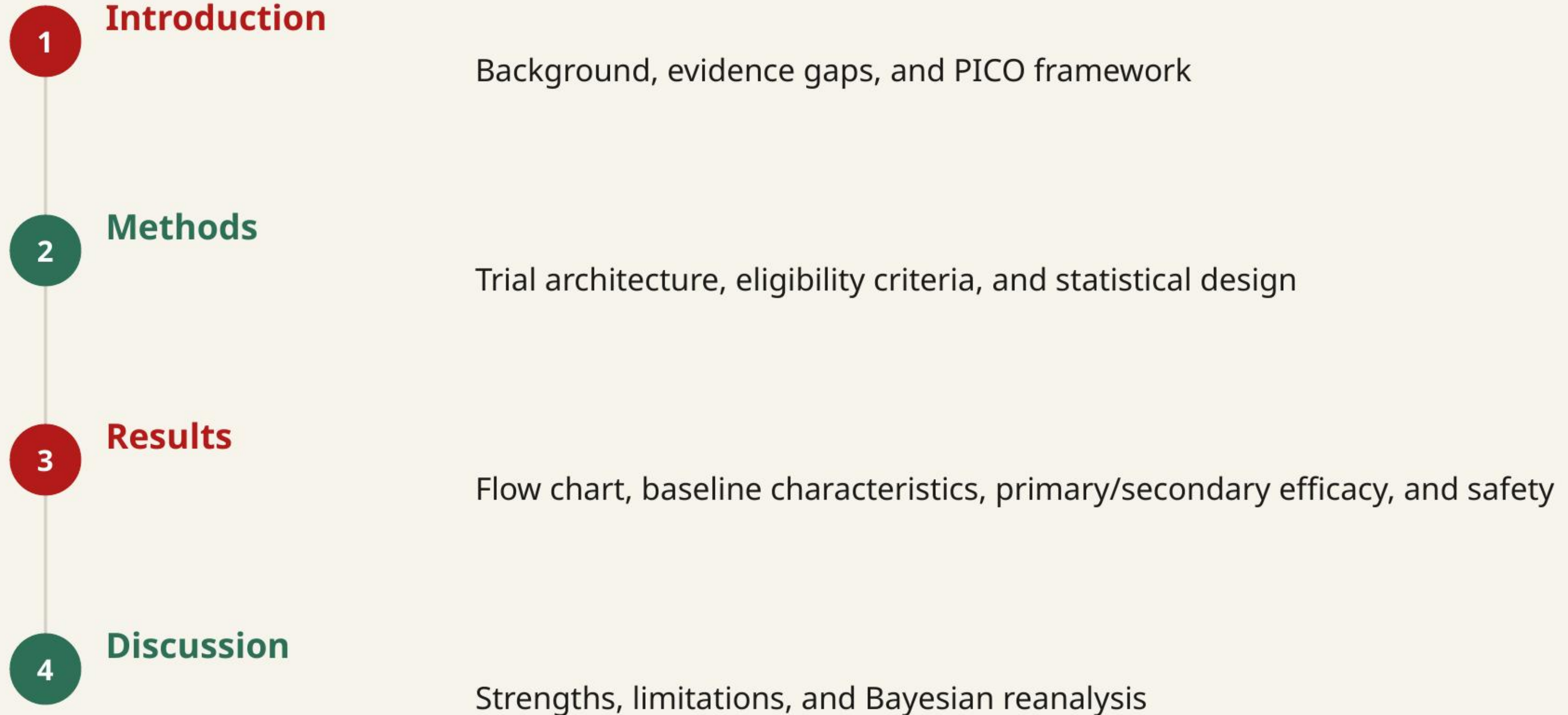
FUNDING & RESEARCH GROUPS

- Full Citation: McGuire DK, Marx N, Mulvagh SL, et al. Oral Semaglutide and Cardiovascular Outcomes in High-Risk Type 2 Diabetes. N Engl J Med 2025;392:2001-2012. doi:10.1056/NEJMoa2501006
- Trial Name: SOUL (Semaglutide Cardiovascular Outcomes Trial).
- Trial Registration: ClinicalTrials.gov number, NCT03914326.

TRIAL REGISTRATION & SPONSORS

- Funding: Funded by Novo Nordisk, who designed, conducted, and analyzed the trial.
- Independent Verification: Data analysis and endpoints were independently verified by Statogen Consulting.
- Sponsors Role: Novo Nordisk was involved in writing the manuscript and decision to submit. Medical writing was funded by the sponsor.

Structure of This Appraisal



INTRODUCTION

Background, Clinical Need, and Trial Rationale

Why This Paper Matters

Clinical problem: Clinical Problem: Type 2 diabetes is associated with a high risk of cardiovascular disease. Approximately 828 million adults are affected worldwide.

Evidence gap: Evidence Gap: Efficacy of injectable semaglutide in reducing MACE and kidney events is well established, but long-term cardiovascular and renal efficacy of the oral formulation was unconfirmed.

Research question: Research Question: Does once-daily oral semaglutide (maximal dose, 14 mg) reduce major cardiovascular events in patients with T2DM and high cardiovascular/renal risk?

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Clinical Question & PICO Framework

- P** — Patients aged 50 years or older with type 2 diabetes (HbA1c 6.5 to 10.0%), and established atherosclerotic cardiovascular disease, chronic kidney disease (eGFR <60), or both.
- I** — Oral semaglutide (maximal dose, 14 mg) once daily, starting at 3 mg and escalated to 7 mg and then 14 mg.
- C** — Matching placebo, both in addition to standard care (glucose-lowering and risk-reducing therapies).
- O** — Primary outcome: major adverse cardiovascular events (MACE: composite of CV death, nonfatal MI, or nonfatal stroke). Follow-up: median 49.5 months.

METHODS

Study Population, Eligibility, Outcomes, and Statistics

Study Design & Methodology

TRIAL ARCHITECTURE

- Design Type: Double-blind, randomized, placebo-controlled, event-driven, superiority phase 3b trial.
- Randomization: 1:1 ratio to oral semaglutide or matching placebo, administered centrally.
- Setting: International, 444 clinical sites in 33 countries across North America, Europe, Asia-Pacific, and Latin America.

BIAS CONTROLS

- Blinding Status: Double-blind (patients, clinicians, and study personnel were fully blinded). Matching placebo tablets were identical.
- Adjudication: Cardiovascular and renal outcome events were assessed by an independent, blinded central adjudication committee.

Eligibility Criteria

KEY INCLUSION CRITERIA

- Age & Diagnosis: Age 50 years or older with type 2 diabetes mellitus.
- Glycemic Control: Glycated hemoglobin (HbA1c) level of 6.5 to 10.0% at screening.
- Risk Profile: Established CAD, cerebrovascular disease, symptomatic peripheral artery disease, or chronic kidney disease (eGFR <60).

KEY EXCLUSION CRITERIA

- End-Stage Kidney Disease: ESKD or patients receiving long-term kidney-replacement therapy were excluded.
- Recent Events: Acute coronary syndrome, stroke, or revascularization within 30-90 days prior to screening were excluded.

Outcomes and Endpoint Hierarchy

- Primary Outcome: Major adverse cardiovascular events (MACE) - composite of death from cardiovascular causes, nonfatal MI, or nonfatal stroke (assessed in time-to-first-event analysis).
- Confirmatory Secondary Outcomes (Hierarchical Testing Order):
 - 1. Major kidney disease events (five-point composite of CV death, renal death, 50% eGFR decline, eGFR <15, or dialysis/transplant).
 - 2. Death from cardiovascular causes.
 - 3. Major adverse limb events (MALE: composite of hospitalization for acute or chronic limb ischemia).
- Supportive Outcomes: Death from any cause, heart failure hospitalization, coronary revascularization, HbA1c and body weight changes.

Analysis Sets and Estimand

- Efficacy Analysis Set: Intention-to-treat (ITT) population, including all randomized participants, regardless of study drug adherence or treatment switches.
- Censoring Rules: Data from participants who withdrew, died, or were lost to follow-up were censored at last contact, death, or withdrawal.
- Missing Data: Handled via Cox proportional-hazards models and verified by tipping-point sensitivity analyses and multiple imputation of event times.
- Safety Analysis Set: All randomized participants who received at least one dose of study treatment.

Statistical Plan & Multiplicity Control

- Sample Size: Estimated 9642 participants to provide 90% power to detect a 17% lower risk of primary MACE (one-sided alpha 0.025, target events ≥ 1225).
- Multiplicity: Strict hierarchical testing chain. Efficacy must be shown at one step to proceed to the next secondary endpoint.
- Alpha spending: Lan-DeMets alpha-spending function used to adjust nominal significance levels for a prespecified interim analysis at 2/3 events.
- Intercurrent Events: Sensitivity analyses adjusted for protocol violations (such as open-label GLP-1 RA use: 3.6% in semaglutide vs 5.2% in placebo).

Methods Critical Appraisal

STRENGTHS

- Double-Blinded Control: Strong blinding compared to open-label trials, with matching placebo and blinded adjudication.
- Standard of Care Integration: Trial permitted concomitant SGLT2 inhibitors (26.9% at baseline) reflecting modern standards.
- Large Sample Size: Robustly powered for primary outcome with high completion rate (98.4% completed the trial).

POTENTIAL FLAWS / RISK OF BIAS

- Industry Sponsor Influence: Trial was designed, conducted, and analyzed by the sponsor (Novo Nordisk), raising potential bias.
- Adherence/Discontinuation: Study drug discontinuation rates were high (27.1% semaglutide vs 28.5% placebo), reflecting real-world GI tolerability.

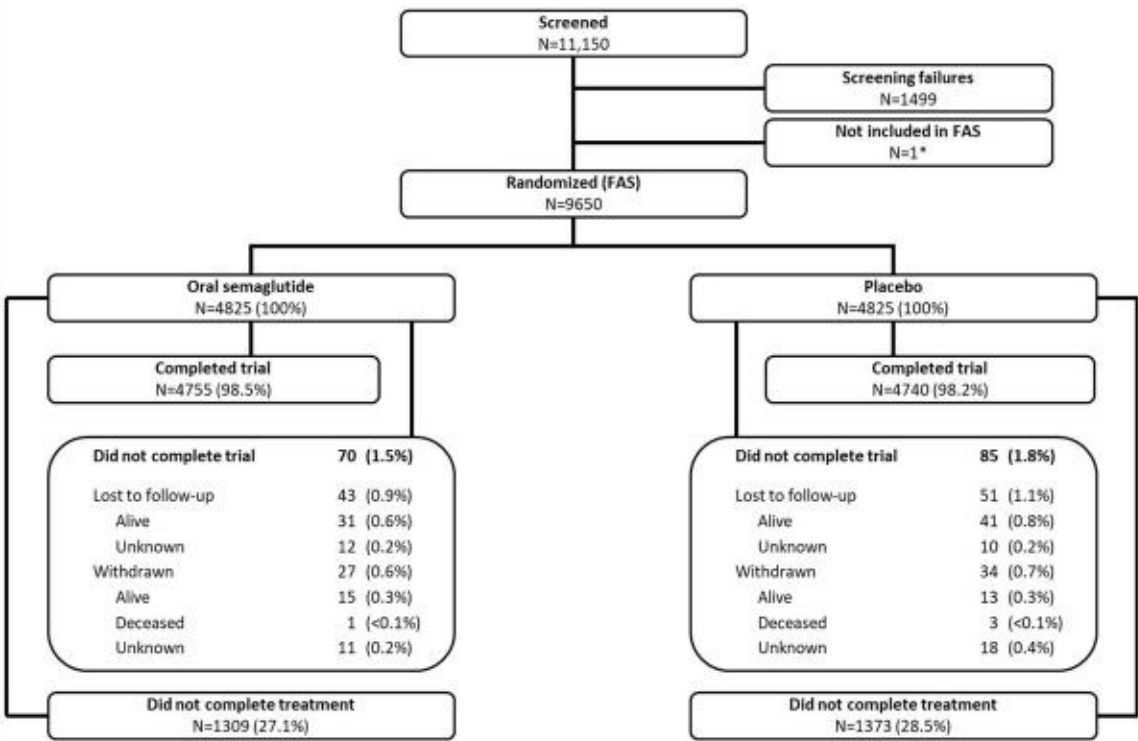
RESULTS

Efficacy Findings, Tables, Figures, and Safety Data

Figure S2. Participant Disposition (CONSORT Flow)

Screening, randomization, dose-escalation, and final follow-up

Figure S2. Disposition of Participants in SOUL



*Randomization was removed due to one participant being randomized more than once.

Participants who completed the trial were defined as those who attended the follow-up visit or died during the trial. Mean duration of exposure to treatment was 42.0±15.5 months overall (41.8±15.9 months for oral semaglutide and 42.1±15.1 months for placebo).

FAS, full analysis set.

Figure S2. Disposition of 9650 randomized participants in the SOUL trial.

Table 1. Characteristics of the Participants at Baseline

Demographics, glycemic indices, comorbidities, and baseline medication

Table 1. Characteristics of the Participants at Baseline.*		
Characteristic	Oral Semaglutide (N = 4825)	Placebo (N = 4825)
Age — yr	66.1±7.6	66.1±7.5
Female sex — no. (%)	1376 (28.5)	1414 (29.3)
Race or ethnic group — no. (%)†		
White	3327 (69.0)	3321 (68.8)
Black	124 (2.6)	128 (2.7)
Asian	1134 (23.5)	1121 (23.2)
American Indian or Alaska Native	7 (0.1)	12 (0.2)
Native Hawaiian or Pacific Islander	4 (<0.1)	5 (0.1)
Other	185 (3.8)	192 (4.0)
Not reported	44 (0.9)	46 (1.0)
Hispanic or Latino ethnic group — no. (%)†	674 (14.0)	706 (14.6)
Body weight — kg	87.5±19.1	88.3±19.6
Body-mass index‡	31.0±5.7	31.2±5.9
Glycated hemoglobin level — mmol/mol	63.6±12.6	63.5±12.3
Glycated hemoglobin level — %	8.0±1.2	8.0±1.1
Median duration of diabetes (IQR) — yr	14.7 (9.0–20.8)	14.6 (8.9–20.8)
History of cardiovascular or kidney disease — no. (%)§		
Cardiovascular disease only	2730 (56.6)	2738 (56.7)
Chronic kidney disease only	632 (13.1)	609 (12.6)
Both cardiovascular and chronic kidney disease	1303 (27.0)	1317 (27.3)
Hypertension — no. (%)	4378 (90.7)	4381 (90.8)
Current smoking — no. (%)	545 (11.3)	584 (12.1)
Systolic blood pressure — mm Hg	134.6±16.3	134.7±16.4
Diastolic blood pressure — mm Hg	76.6±10.1	76.7±10.1
Pulse — beats/min	72.8±11.1	72.9±11.4
Median high-sensitivity C-reactive protein level (IQR) — mg/liter	2.0 (0.9–4.3)	2.0 (0.9–4.5)
eGFR — ml/min/1.73 m²¶	74.0±22.6	73.6±22.6

* Plus-minus values are means ±SD. Percentages may not total 100 because of rounding. A full description of participant characteristics and medications used at baseline is provided in Table S1. The abbreviation eGFR denotes estimated glomerular filtration rate, and IQR interquartile range.

† Race and ethnic group were reported by the participant.

‡ The body-mass index is the weight in kilograms divided by the square of the height in meters.

§ For 3.3% of the participants, whether only one criterion or two criteria were fulfilled was unknown. Chronic kidney disease was defined by an eGFR of less than 60 ml per minute per 1.73 m²; the most recent eGFR available in the medical record was used if it had been obtained within the previous 6 months.

¶ The eGFR was measured at randomization with the use of the Chronic Kidney Disease Epidemiology Collaboration method.¹²

Among the five components of this composite outcome, death from cardiovascular causes accounted for 71.2% of the events, whereas 28.8% were kidney-related events. The remaining two confirmatory secondary outcomes in the hierarchy were not tested for significance: death from cardiovascular causes (hazard ratio, 0.93; 95% CI, 0.80 to 1.09) (Fig. 1B) and major adverse limb events (hazard ratio, 0.71; 95% CI, 0.52 to 0.96) (Fig. 1F).

SUPPORTIVE SECONDARY OUTCOMES

The results for additional efficacy outcomes are summarized in Table 2. The hazard ratio (oral

Baseline Comparability Appraisal

- Demographic Balance: Baseline characteristics were highly balanced. Mean age 66.1 years, 28.9% women, and 23.5% Asian.
- Comorbidity Profiles: High-risk cohort: 70.7% had coronary artery disease, 23.1% had heart failure, and 42.4% had chronic kidney disease.
- Baseline Glycemia: Mean HbA1c was 8.0% in both groups, and median diabetes duration was 14.7 years.
- Concomitant Therapies: Well-matched: 26.9% in both groups were on SGLT2 inhibitors at baseline, and 90.7% had hypertension.
- Clinical Relevance: No clinically meaningful imbalances. The population represents a very high-risk secondary prevention cohort.

Table 2. Primary and Secondary Efficacy Outcomes

Detailed event rates, incidence per 100 person-years, and hazard ratios

Table 2. Primary and Secondary Efficacy Outcomes.*						
Outcome	Oral Semaglutide (N = 4825)		Placebo (N = 4825)		Hazard Ratio (95% CI)	P Value†
	no. of participants with event (%)	no. of events per 100 person- yr	no. of participants with event (%)	no. of events per 100 person- yr		
Primary outcome						
Major adverse cardiovascular events, three-point composite‡	579 (12.0)	3.1	668 (13.8)	3.7	0.86 (0.77–0.96)	0.006
Confirmatory secondary outcomes						
Major kidney disease events, five-point composite§	403 (8.4)	2.1	435 (9.0)	2.3	0.91 (0.80–1.05)	0.19
Death from cardiovascular causes	301 (6.2)	1.6	320 (6.6)	1.7	0.93 (0.80–1.09)	—
Major adverse limb events, two-point composite¶	71 (1.5)	0.4	99 (2.1)	0.5	0.71 (0.52–0.96)	—
Supportive secondary outcomes						
Major adverse cardiovascular events, five-point composite	670 (13.9)	3.6	777 (16.1)	4.3	0.84 (0.76–0.93)	—
Nonfatal myocardial infarction	191 (4.0)	1.0	253 (5.2)	1.4	0.74 (0.61–0.89)	—
Fatal or nonfatal myocardial infarction	200 (4.1)	1.1	268 (5.6)	1.4	0.73 (0.61–0.88)	—
Nonfatal stroke	144 (3.0)	0.8	161 (3.3)	0.9	0.88 (0.70–1.11)	—
Fatal or nonfatal stroke	164 (3.4)	0.9	171 (3.5)	0.9	0.95 (0.76–1.17)	—
Coronary revascularization	200 (4.1)	1.1	263 (5.5)	1.4	0.75 (0.62–0.90)	—
Hospitalization for unstable angina pectoris	74 (1.5)	0.4	80 (1.7)	0.4	0.92 (0.67–1.26)	—
Death from any cause	528 (10.9)	2.8	577 (12.0)	3.0	0.91 (0.80–1.02)	—
Death from noncardiovascular causes	227 (4.7)	1.2	257 (5.3)	1.4	0.87 (0.73–1.04)	—
Heart failure events, three-point composite**	405 (8.4)	2.1	443 (9.2)	2.4	0.90 (0.79–1.03)	—
Heart failure	146 (3.0)	0.8	167 (3.5)	0.9	0.86 (0.69–1.08)	—
Major kidney disease events, four-point composite††	112 (2.3)	0.6	129 (2.7)	0.7	0.86 (0.66–1.10)	—
Death from kidney-related causes	1 (<0.1)	<0.1	7 (0.1)	<0.1	0.14 (0.01–0.79)	—
Severe hypoglycemic episode	76 (1.6)	0.5	84 (1.7)	0.6	0.90 (0.66–1.22)	—
* Time-to-first-event outcomes are shown for the intention-to-treat population (all the individual participants who had undergone randomization) during the trial observation period. All the outcomes were analyzed with the use of a Cox proportional-hazards model with randomized group assignment as a categorical fixed factor. Confidence intervals for supportive secondary outcomes were not adjusted for multiplicity and therefore cannot be used for hypothesis testing. Data from participants without events of interest were censored at the end of their trial observation period.						
† Two-sided P values are shown. After accounting for the results from the interim analysis, the nominal two-sided significance level for the primary outcome was 0.04561. The nominal two-sided significance level for the first confirmatory secondary outcome was 0.04433.						
‡ As the primary outcome, the major adverse cardiovascular events outcome is a three-point composite of death from cardiovascular causes, nonfatal myocardial infarction, or nonfatal stroke.						
§ As the first confirmatory secondary outcome, the major kidney disease events outcome is a five-point composite of death from cardiovascular causes, death from kidney-related causes, a persistent reduction from baseline in the eGFR of 50% or more as measured with the Chronic Kidney Disease Epidemiology Collaboration method, a persistent eGFR of less than 15 ml per minute per 1.73 m ² , or the initiation of long-term kidney-replacement therapy with either dialysis or transplantation.						
¶ The major adverse limb events outcome is a two-point composite of hospitalization for acute limb ischemia or hospitalization for chronic limb ischemia.						
As a supportive secondary outcome, the major adverse cardiovascular events outcome is a five-point composite of death from cardiovascular causes, nonfatal myocardial infarction, nonfatal stroke, coronary revascularization, or hospitalization for unstable angina.						
** The heart failure events outcome is a three-point composite of death from cardiovascular causes, an urgent visit for heart failure, or hospitalization for heart failure.						
†† As a supportive secondary outcome, the major kidney disease events outcome is a four-point composite of death from kidney-related causes, a persistent reduction from baseline in the eGFR of 50% or more as measured with the Chronic Kidney Disease Epidemiology Collaboration method, a persistent eGFR of less than 15 ml per minute per 1.73 m ² , or the initiation of long-term kidney-replacement therapy with either dialysis or transplantation.						

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Table 2. Detailed event rates, incidence per 100 person-years, hazard ratios, and confidence intervals.

Figure 1A & 1B. Primary MACE and CV Death

Oral semaglutide vs placebo cumulative incidence curves

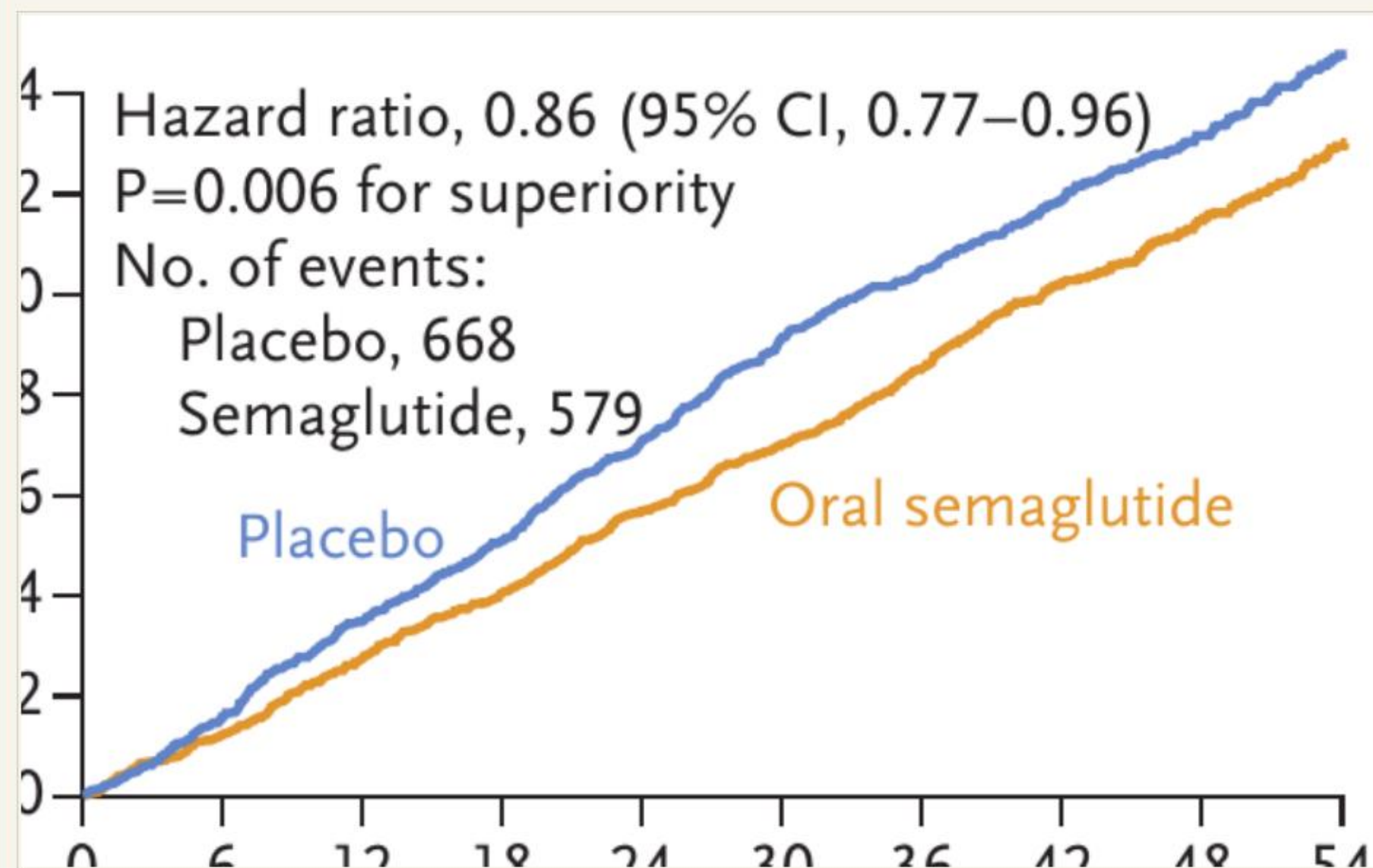


Figure 1A. Primary MACE Outcome

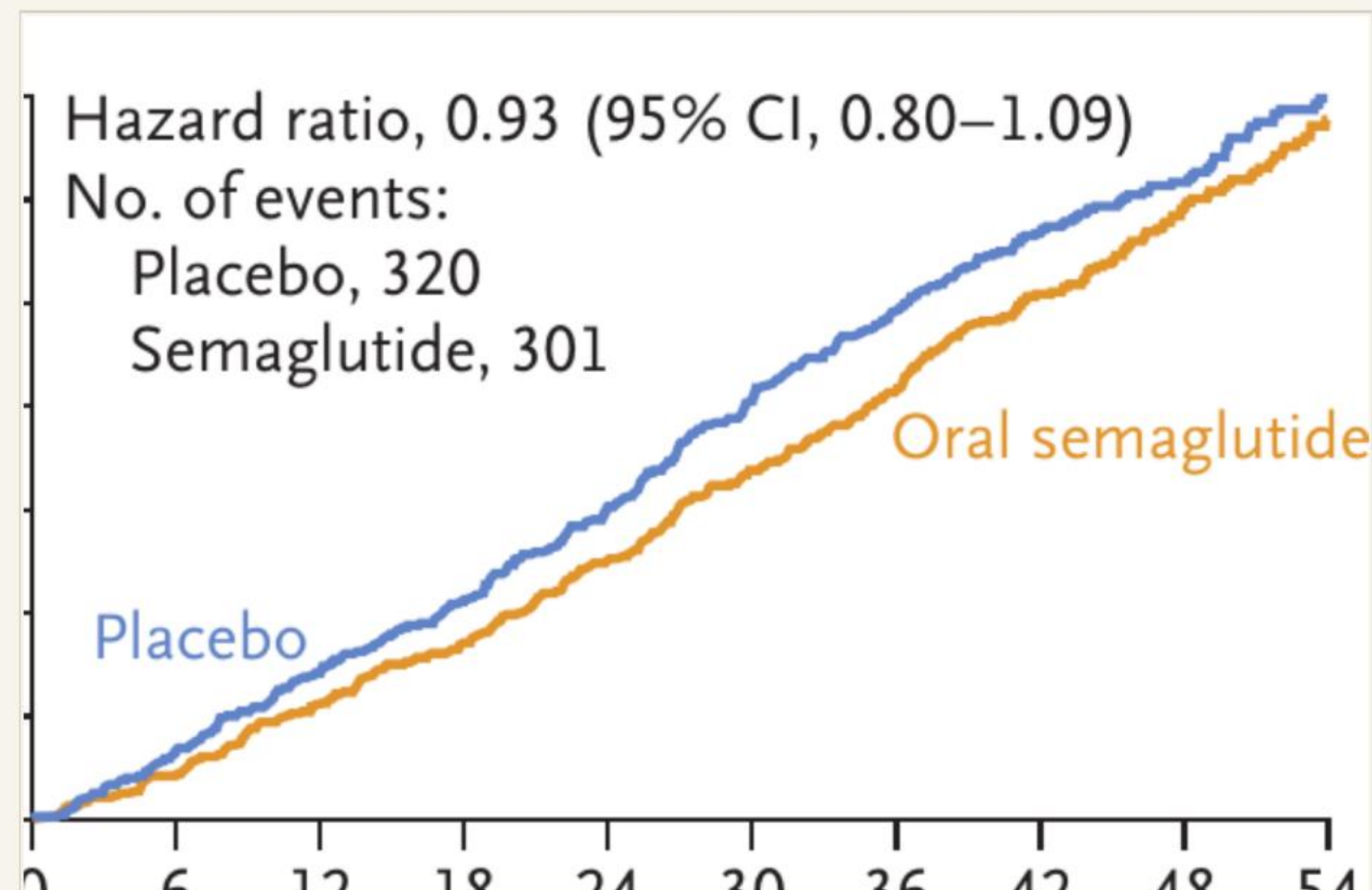


Figure 1B. Death from Cardiovascular Causes

Figure 1C & 1D. Nonfatal MI and Nonfatal Stroke

Oral semaglutide vs placebo cumulative incidence curves

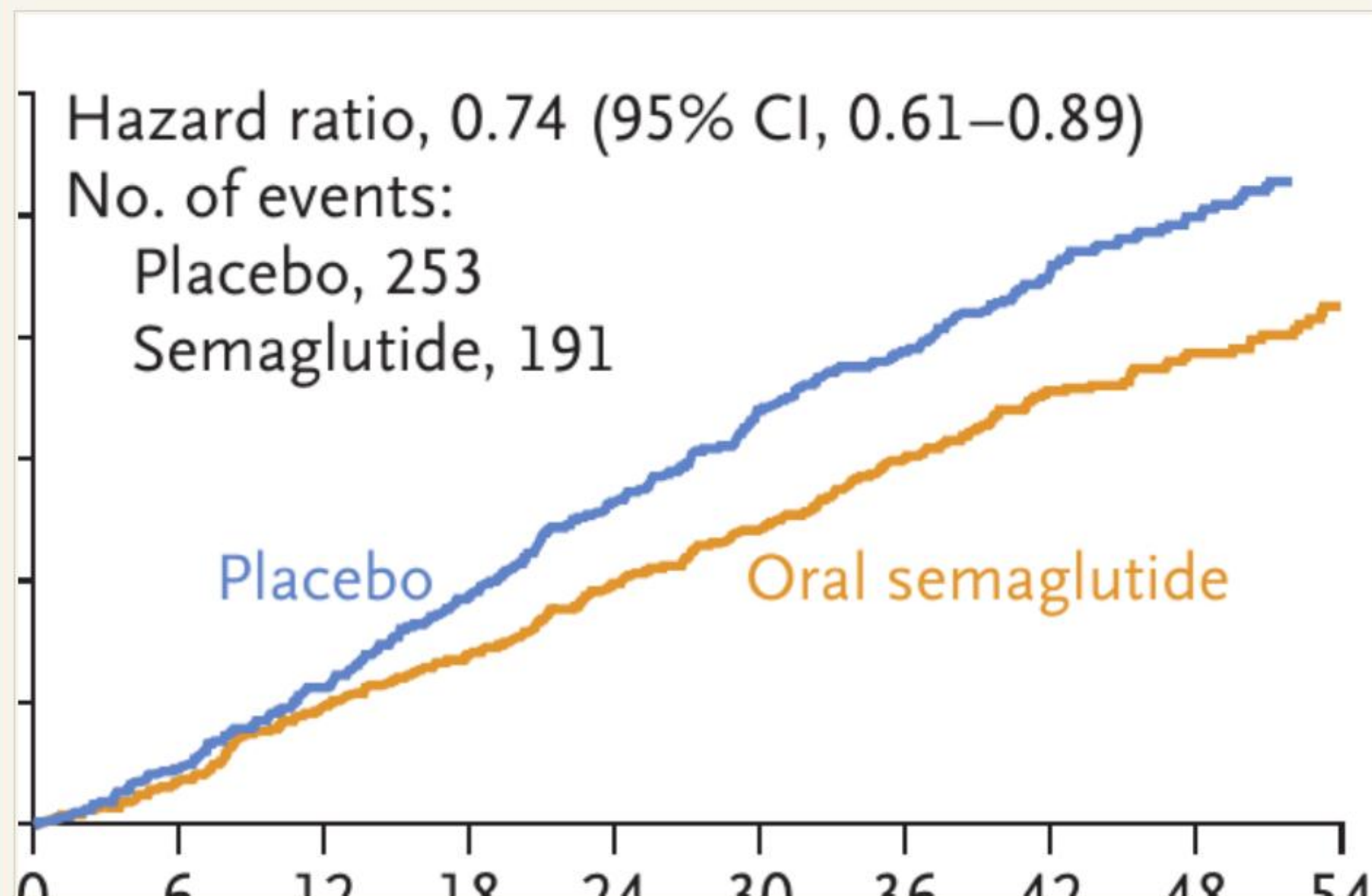


Figure 1C. Nonfatal Myocardial Infarction

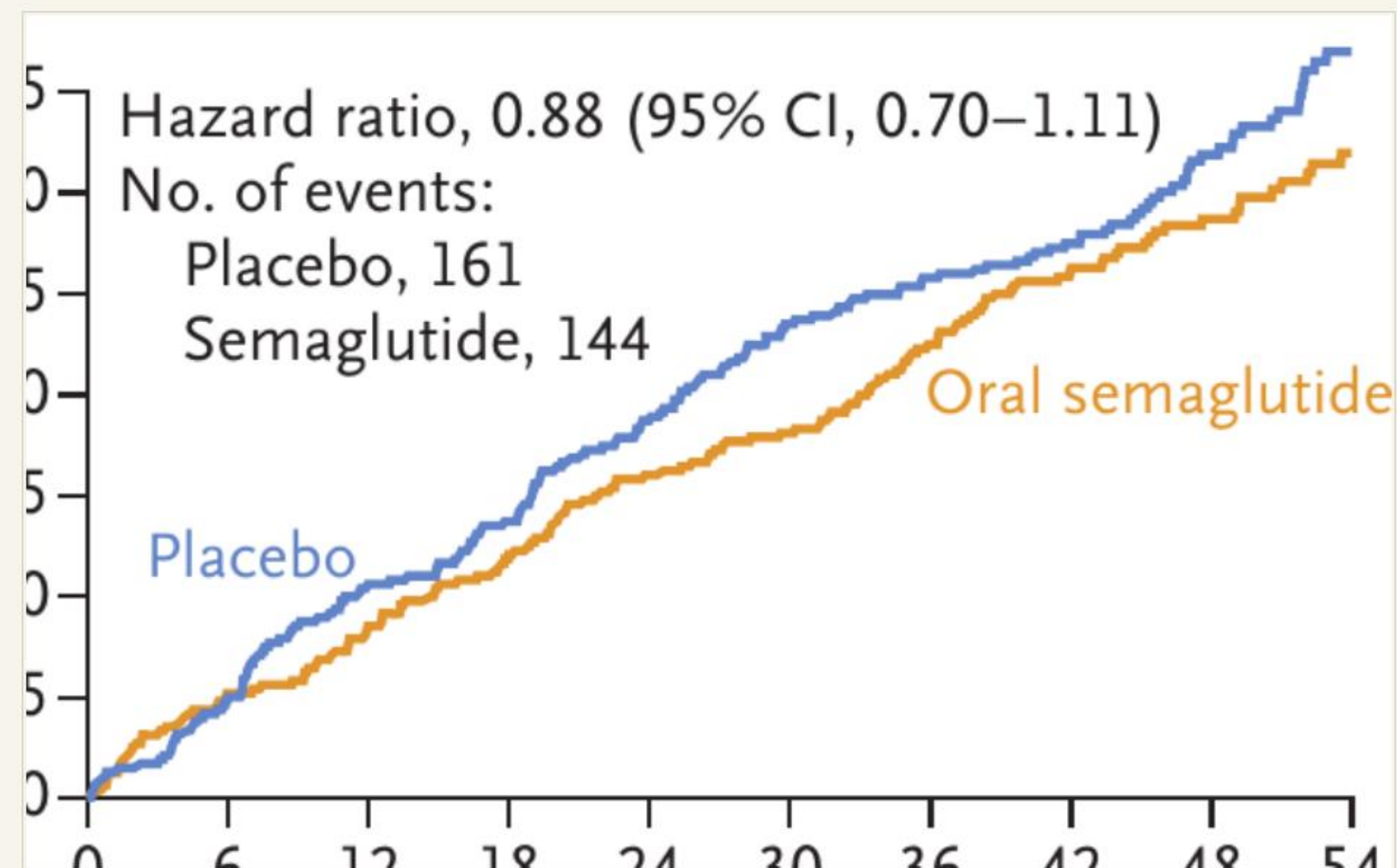


Figure 1D. Nonfatal Stroke

Figure 1E & 1F. Major Kidney Events and Limb Events

Oral semaglutide vs placebo cumulative incidence curves

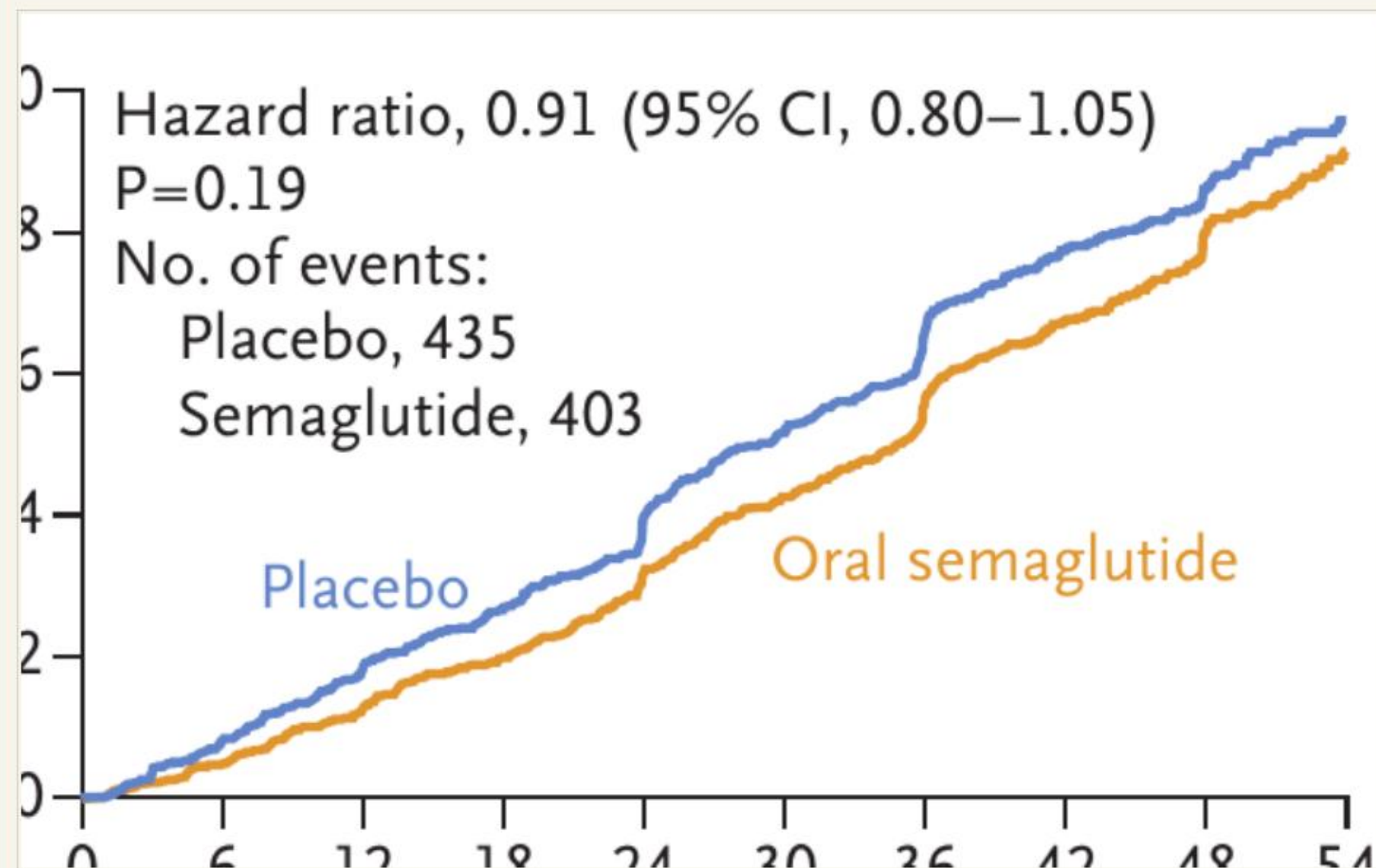


Figure 1E. Major Kidney Disease Events

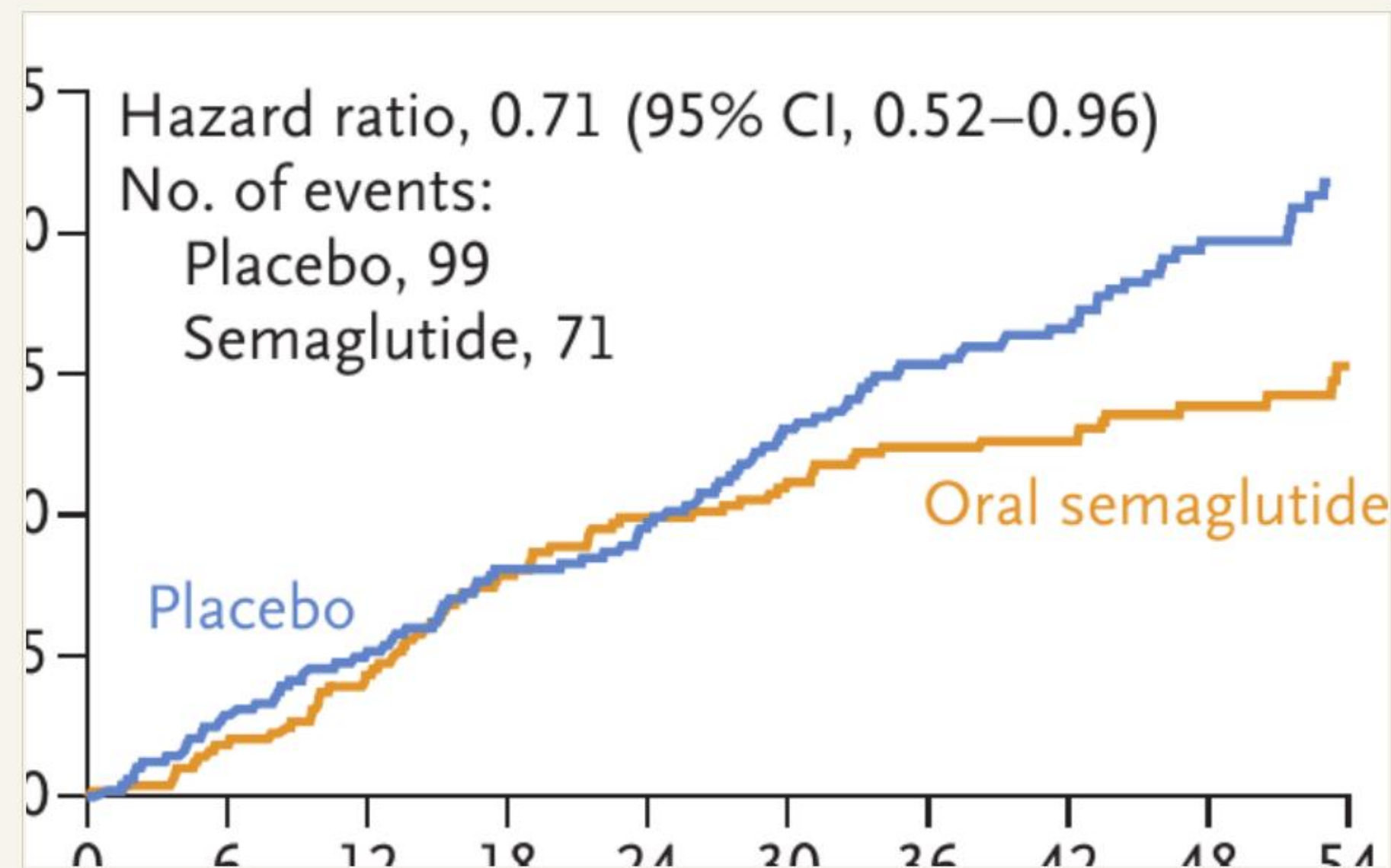
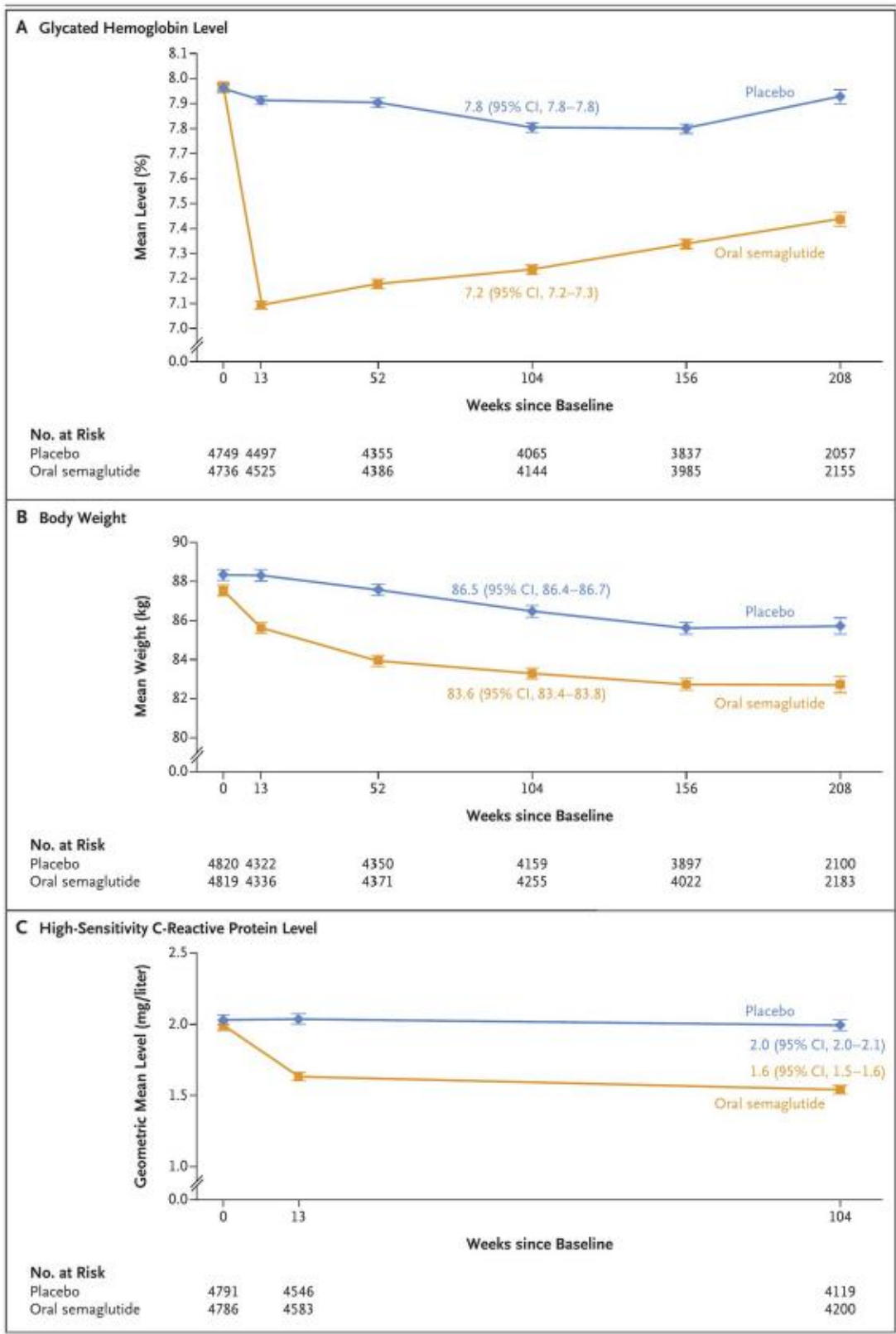


Figure 1F. Major Adverse Limb Events

Figure 2. Changes in HbA1c, Body Weight, and CRP

Observed mean changes over 104 weeks in the ITT population



2010

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Figure 2. Mean changes in HbA1c (-0.56% difference), body weight (-2.95 kg difference), and hsCRP over 104 weeks.

Efficacy Summary: Relative & Absolute Effects

- Primary Outcome Benefit: MACE was significantly reduced by 14% (12.0% vs 13.8%; HR 0.86; 95% CI, 0.77-0.96; P=0.006).
- Absolute Risk Reduction (ARR): 1.84 percentage points (13.84% vs 12.00%) over a median of 49.5 months.
- Number Needed to Treat (NNT): 54 patients treated for 4 years to prevent 1 MACE event.
- Secondary Efficacy Hierarchy: Testing stopped at the first secondary outcome: Major kidney disease events was neutral (8.4% vs 9.0%; HR 0.91; 95% CI, 0.80-1.05; P=0.19).
- Exploratory Efficacy: Nonfatal MI was reduced by 26% (4.0% vs 5.2%; HR 0.74; 95% CI, 0.61-0.89); Major adverse limb events was reduced by 29% (HR 0.71; 95% CI, 0.52-0.96).

Table S4. Safety and Adverse Events Summary

Serious adverse events (SAEs) and treatment discontinuations

Table S4. Adverse Events Summary			
Event	Oral semaglutide (N = 4825)	Placebo (N = 4825)	P value
	Number of participants (%)		
Serious adverse events* — n (%)	2312 (47.9)	2427 (50.3)	0.02
Cardiac disorders	861 (17.8)	954 (19.8)	0.02
Infections and infestations	726 (15.0)	797 (16.5)	0.05
Nervous system disorders	382 (7.9)	387 (8.0)	0.88
Neoplasms benign, malignant, and unspecified	330 (6.8)	274 (5.7)	0.02
Renal and urinary disorders	249 (5.2)	287 (5.9)	0.10
Gastrointestinal disorders	239 (5.0)	210 (4.4)	0.18
Vascular disorders	237 (4.9)	222 (4.6)	0.50
Injury, poisoning and procedural complications	218 (4.5)	304 (6.3)	<0.001
General disorders and administration-site conditions	206 (4.3)	184 (3.8)	0.28
Respiratory, thoracic, and mediastinal disorders	185 (3.8)	217 (4.5)	0.11
Metabolism and nutrition disorders	182 (3.8)	219 (4.5)	0.07
Musculoskeletal and connective tissue disorders	161 (3.3)	169 (3.5)	0.70
Hepatobiliary disorders	115 (2.4)	101 (2.1)	0.37
Eye disorders	102 (2.1)	97 (2.0)	0.77
Adverse events leading to permanent discontinuation of trial product, regardless of severity† — n (%)	749 (15.5)	559 (11.6)	<0.001
Gastrointestinal disorders	310 (6.4)	98 (2.0)	<0.001
Neoplasms benign, malignant, and unspecified	79 (1.6)	61 (1.3)	0.15
Infections and infestations	63 (1.3)	96 (2.0)	0.01
Nervous system disorders	63 (1.3)	61 (1.3)	0.93
Cardiac disorders	59 (1.2)	74 (1.5)	0.22
Renal and urinary disorders	29 (0.6)	31 (0.6)	0.90

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Table S4. Summary of serious adverse events (SAEs) and events leading to discontinuation.

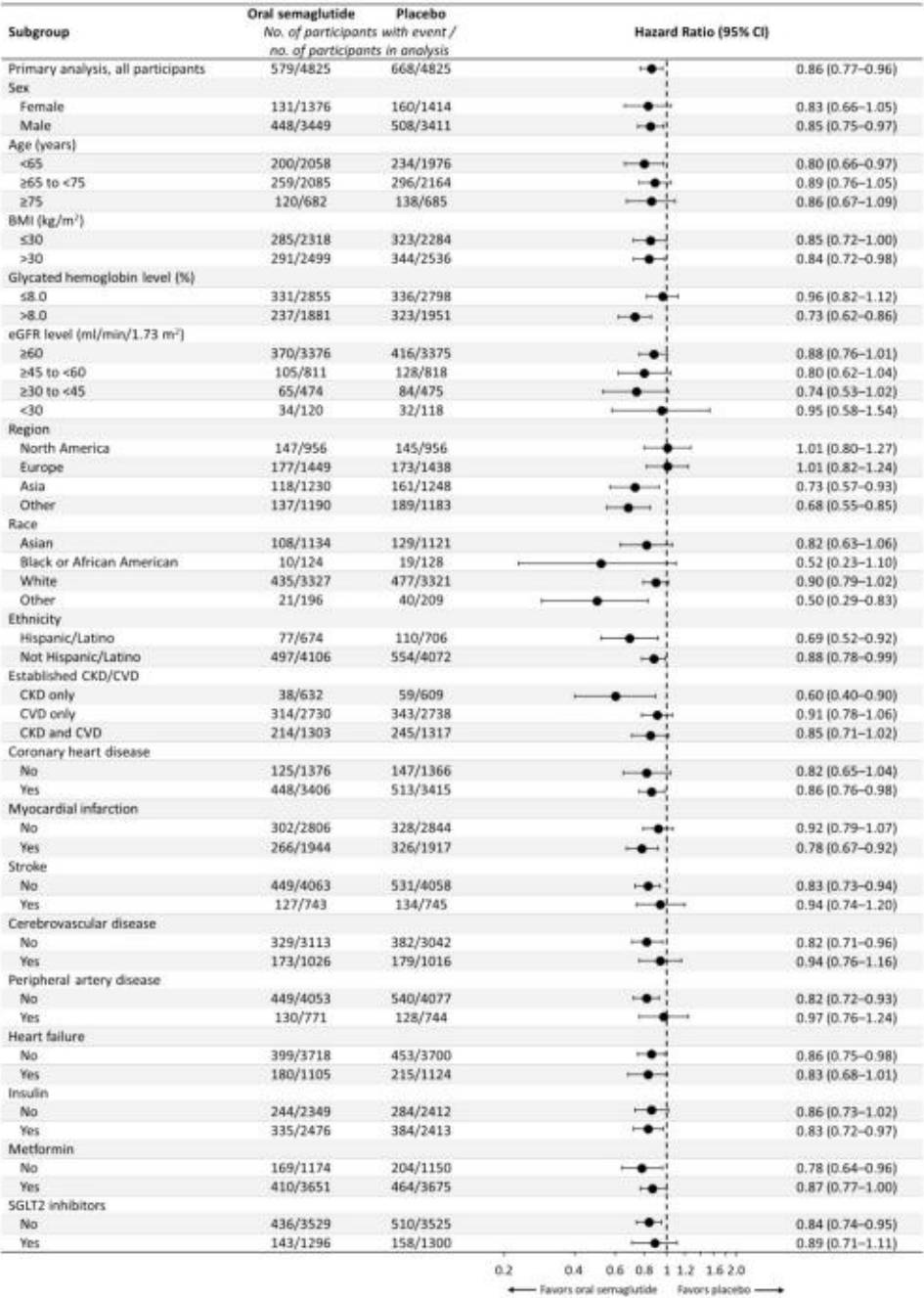
Safety Summary: Tolerability & Adverse Events

- Serious Adverse Events: Lower in the oral semaglutide group (47.9%) than the placebo group (50.3%) (P=0.02). Primarily driven by fewer cardiac disorders in the drug group.
- Treatment Discontinuations: Significantly higher in the oral semaglutide group (15.5%) than the placebo group (11.6%), primarily due to gastrointestinal disorders (6.4% vs 2.0%).
- Gastrointestinal Side Effects: Nausea, vomiting, diarrhea, and constipation were more common in the oral semaglutide group.
- Pancreatitis & Malignancy: Acute pancreatitis occurred in 0.4% in both groups (no increase). Malignant neoplasms were balanced (balanced across groups).
- Hypoglycemia: Severe hypoglycemia was lower in the semaglutide group than placebo (1.6% vs 1.7%, 88 vs 121 episodes).

Figure S4. Subgroup Analysis of the Primary Outcome

Consistency of the primary MACE reduction across baseline characteristics

Figure S4. Subgroup Analysis of the Primary Outcome



Subgroups were defined by values at baseline. CKD was defined as eGFR <60 ml/min/1.73 m².

For the primary analysis, the hazard ratios and CIs were adjusted for the group sequential design using the likelihood ratio ordering. For the subgroup analyses, estimated hazard ratios and corresponding CIs were estimated using a Cox proportional hazards model with interaction between treatment group and the relevant subgroup as fixed factor.

BMI, body mass index; CI, confidence interval; CKD, chronic kidney disease; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; SGLT2i, sodium-glucose cotransporter-2 inhibitor.

Figure S4. Primary MACE outcome across prespecified baseline subgroups.

Results Critical Appraisal

- MACE Benefit Robustness: The primary MACE outcome was robustly positive and consistent across all sensitivity analyses.
- Neutral Renal Efficacy: Contrast with FLOW trial (injectable semaglutide showed 24% renal MACE reduction). Likely due to higher baseline eGFR (74 vs 47) and lower subcutaneous bioavailability of oral formulation.
- Exploratory Endpoint Caution: Efficacy benefits in limb events (MALE) and MI are technically exploratory due to the neutral renal composite endpoint in the hierarchy.
- Real-world Adherence: High permanent discontinuation (27.1%) in clinical trials means real-world clinical benefits may be lower due to poorer adherence.

Risk of Bias 2 (RoB 2) Summary

OVERALL RISK OF BIAS JUDGEMENT

LOW RISK OF BIAS

GRADE Certainty of Evidence

GRADE FINAL CERTAINTY

HIGH CERTAINTY

DISCUSSION

Bayesian Reanalysis, Key Trial Context, and Guideline Consensus

Chronology of Key GLP-1 Receptor Agonist Trials

Comparison of SOUL with historical cardiovascular outcome trials

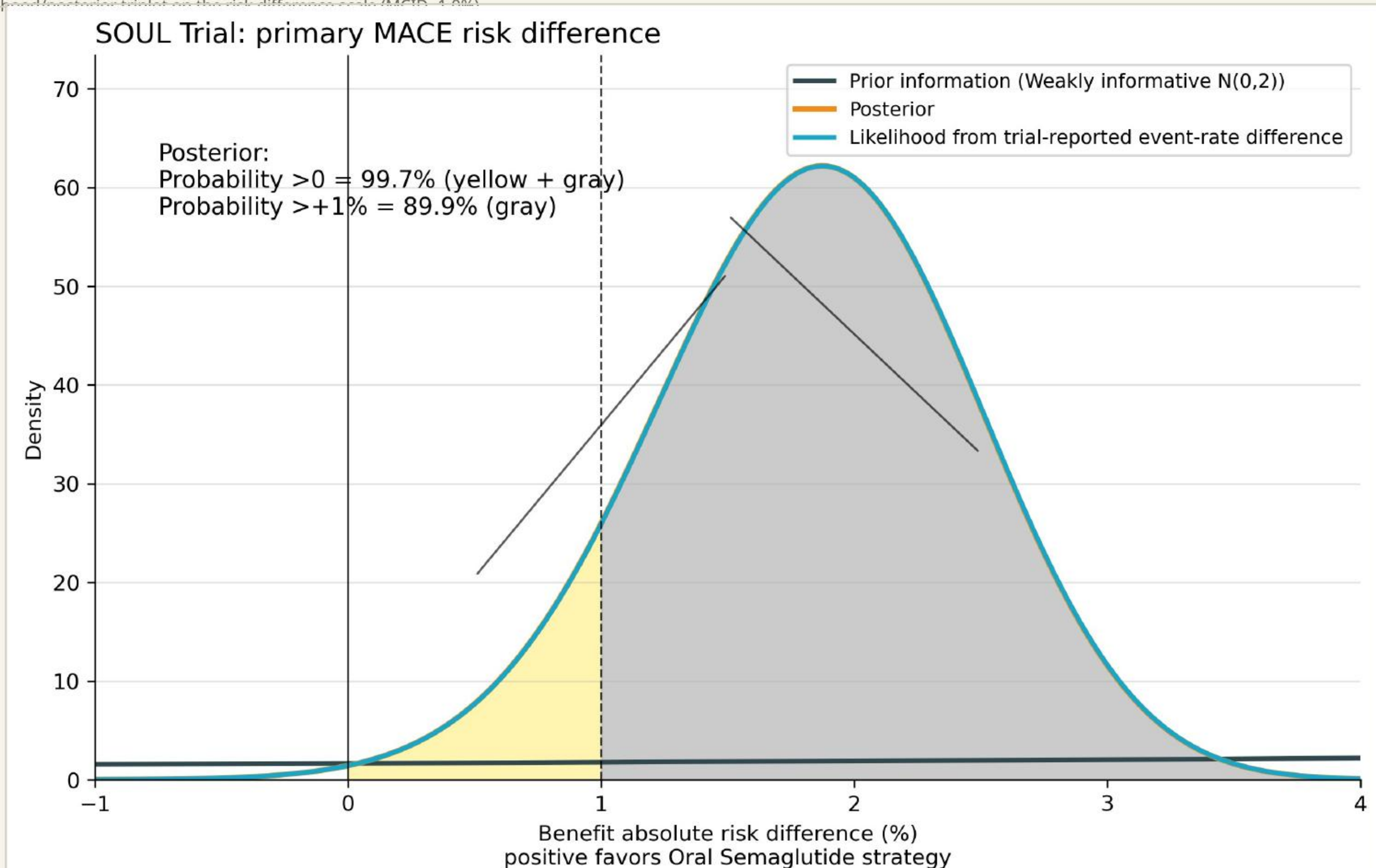
Trial	Drug & Dose	Population (N)	Follow-up	MACE HR (95% CI)
LEADER (2016)	Liraglutide 1.8 mg SC QD	T2DM + High CV risk (9,340)	3.8 years	0.87 (0.78 - 0.97)
SUSTAIN-6 (2016)	Semaglutide 0.5/1.0 mg SC QW	T2DM + High CV risk (3,297)	2.1 years	0.74 (0.58 - 0.95)
PIONEER-6 (2019)	Oral Semaglutide 14 mg QD	T2DM + High CV risk (3,183)	1.3 years	0.79 (0.57 - 1.11)
FLOW (2024)	Semaglutide 1.0 mg SC QW	T2DM + CKD (3,533)	3.4 years	0.82 (0.72 - 0.94)
SOUL (2025)	Oral Semaglutide 14 mg QD	T2DM + CKD/CVD (9,650)	4.1 years	0.86 (0.77 - 0.96)

Current Guideline Consensus on GLP-1 RAs

- ADA Standards of Care: Recommends GLP-1 RAs with proven CV benefit as first-line or adjunct therapy for patients with T2DM and established ASCVD or high CV risk, independent of HbA1c.
- KDIGO 2023 Guidelines: Recommends SGLT2 inhibitors as first-line, with GLP-1 RAs as preferred second-line for patients with CKD and T2DM who do not meet glycemic targets.
- ESC 2023 Guidelines: Class I recommendation for GLP-1 RAs in T2DM with established CVD to reduce cardiovascular events, independent of baseline HbA1c.
- Oral vs Injectable Preference: Guidelines support patient choice, noting oral formulation may improve early acceptance and adherence compared to weekly injections.

Posterior Distribution of Risk Difference (Weakly Informative Prior)

Conjugate prior/likelihood



Conjugate normal-normal triplot update based on SOUL trial data. Under a weakly informative prior, the posterior mean absolute risk reduction (RD) is 1.84% (95% CrI, 0.53% to 3.05%), and the posterior probability of exceeding the 1.0% MCID is 89.9%.

Posterior Efficacy under Alternative Priors

Prior sensitivity analysis on MACE absolute risk reduction (RD) scale

Prior Profile	Prior Mean (SD) *	Posterior Mean RD (95% CrI)	P(RD > 0)	P(RD > 1.0% MCID)
Noninformative (Default)	0.00 (10.00)	1.84% (0.53% to 3.05%)	99.7%	89.9%
Weakly Informative	0.00 (2.00)	1.84% (0.53% to 3.05%)	99.7%	89.9%
Neutral Moderate	0.00 (1.00)	1.84% (0.52% to 3.04%)	99.6%	89.8%
Neutral Conservative	0.00 (0.50)	1.82% (0.51% to 3.02%)	99.6%	89.4%
Enthusiastic	0.40 (0.50)	1.88% (0.58% to 3.07%)	99.7%	91.0%
Pessimistic	-0.20 (0.50)	1.79% (0.48% to 2.99%)	99.6%	88.5%
Skeptical	0.00 (0.25)	1.75% (0.46% to 2.93%)	99.6%	87.7%
Strong Skeptical	0.00 (0.20)	1.70% (0.43% to 2.86%)	99.5%	86.3%

* Note: Prior mean and SD are specified on the log-odds-ratio scale. Risk differences (RD) are drug minus placebo (positive values favor Oral Semaglutide). MCID is 1.0% absolute risk difference.

APPLICABILITY & TAKEAWAYS

Clinical Significance, Local Patient Context, and Bedside Takeaways

Applicability to North Indian Patients

OPD / CLINICAL FEASIBILITY

- Patient Preference: High preference for oral drugs over subcutaneous injections in Ghaziabad and North Indian settings.
- Dosing Rigor: Correct administration is critical. Take empty stomach in the morning with up to 120 ml water and wait 30 minutes before eating.
- Comorbidity Match: High burden of CAD and CKD in North Indian diabetic patients matches the study population profile.

SOCIOECONOMIC & OTHER BARRIERS

- Cost Barriers: Oral semaglutide is highly expensive. Cost-effectiveness is a major hurdle for average patients.
- Dietary & GI Tolerability: High prevalence of dyspepsia and GI issues in the local population. Requires careful dose titration.

Clinical Takeaways for Hospital Practice

- OPD Practice: Excellent first-line or add-on choice for diabetic patients with established CAD or early CKD (eGFR >30) who prefer oral medication. Ensure careful titration (3 mg -> 7 mg -> 14 mg).
- Wards Practice: Initiate during discharge planning for diabetic patients admitted with myocardial infarction or limb ischemia once stable and tolerating oral intake.
- ICU/HDU Practice: Avoid initiating during acute critical illness, diabetic ketoacidosis, or in patients with gastroparesis, active pancreatitis, or bowel obstruction.
- Concomitant SGLT2i: Efficacy was additive. Do not stop SGLT2 inhibitors to start oral semaglutide; combine them in patients with both CAD and CKD if tolerated and affordable.

Journal Club Discussion Questions

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